

Graph on the Coordinate Plane

Flagship

Lesson 9-1-flagship

Name: _____

Date: _____

Class: _____

LOST COVE MISSION

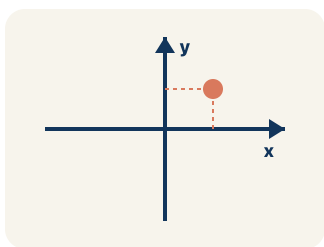
Captain Vega's Coordinates

You are the navigator on Captain Vega's expedition. Instead of 'X marks the spot,' she hid the treasure route as ordered pairs scrawled across a four-quadrant chart. Plot every point in the right order and the lost cove treasure is yours.

Key Vocabulary

Level 2 Standard

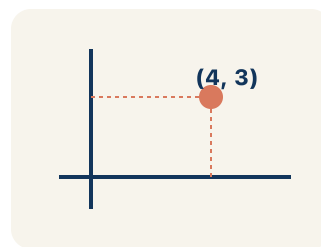
Picture first, then the word, then a plain-language meaning. Say each word out loud.



A grid with a horizontal line (x-axis) crossing a vertical line (y-axis), making four sections

Coordinate plane

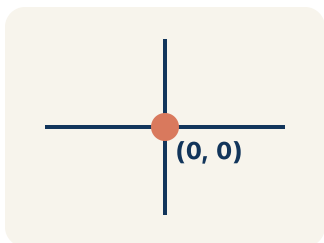
Write the definition:



(3, 5) means move right 3 from the origin, then up 5

Ordered pair

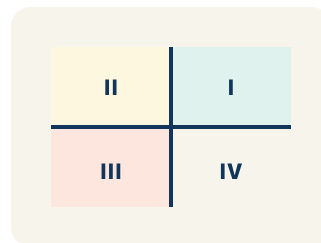
Write the definition:



The starting point at the center where both axes cross — (0, 0)

Origin

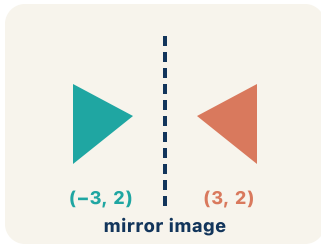
Write the definition:



I (+,+) top-right, II (-,+) top-left, III (-,-) bottom-left, IV (+,-) bottom-right

Quadrant

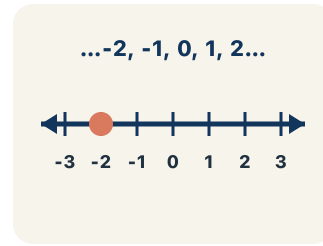
Write the definition:



*$(3, 2)$ reflected over the x -axis becomes $(3, -2)$ —
same x , opposite y*

Reflection

Write the definition:



Integer

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can plot and identify points on the coordinate plane using ordered pairs.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Coordinate plane

Ordered pair

Origin

Quadrant

Reflection

Integer

1 The grid made by a horizontal and a vertical number line is the _____.

2 Two numbers (x, y) that tell the location of a point are an _____.

3 The point (0, 0) where the x-axis and y-axis cross is the _____.

4 One of the four regions of the coordinate plane is a _____.

5 A flip of a point or shape over an axis is a _____.

6 A whole number or its opposite, with no fraction part, is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which ordered pair names a point 3 units right and 7 units up from the origin?

- A. (3, 7)
- B. (7, 3)
- C. (3, 3)
- D. (0, 7)

Step 1 The first coordinate is horizontal (right 3), the second is vertical (up 7).

Step 2 So (3, 7).

 **Answer:** A. (3, 7)

 We do — together

Solve this one with your class using the same steps.

Problem: What are the coordinates of the origin?

- A. (0, 0)
- B. (1, 1)
- C. (0, 1)
- D. (1, 0)

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: A point is at (6, 2). What does the 6 tell you?

- A. Move 6 units to the right
- B. Move 6 units up
- C. Move 6 units to the left
- D. The point is in Quadrant 6

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Stage 2 — The next clue points to Skull Rock at $(-5, 3)$. Which quadrant is Skull Rock in?

- A. Quadrant II
- B. Quadrant I
- C. Quadrant III
- D. Quadrant IV

Show your work:

2. Stage 5 — Captain Vega marks the final treasure 0 units across and 8 units up from the origin. Where does this point land?

- A. On the y-axis at $(0, 8)$
- B. On the x-axis at $(8, 0)$
- C. In Quadrant I at $(8, 8)$
- D. At the origin $(0, 0)$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says that $(3, 5)$ and $(5, 3)$ are the same point because they use the same numbers. Is the student correct? Explain why or why not, and describe where each point is located on the coordinate plane.

Sentence starter: The student is ____ because $(3, 5)$ means ____ while $(5, 3)$ means ____ . The ORDER matters because ____ .

Show your work:

Reflect — Exit Ticket

Point P is located 6 units right and 3 units up from the origin. What ordered pair names point P?

- A. $(6, 3)$
- B. $(3, 6)$
- C. $(6, 0)$
- D. $(0, 3)$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Coordinate plane
2. **Notes 2:** Ordered pair
3. **Notes 3:** Origin
4. **Notes 4:** Quadrant
5. **Notes 5:** Reflection
6. **Notes 6:** Integer
7. **We Try (notes):** A. $(0, 0)$ — *The origin is where the x-axis and y-axis cross, at $(0, 0)$.*
8. **You Try (notes):** A. Move 6 units to the right — *The first number (x-coordinate) tells you horizontal movement. 6 means go right 6 units from the origin.*
9. **Try It 1:** A. Quadrant II — *The x-coordinate is negative (left) and the y-coordinate is positive (up). Negative x and positive y land in the top-left region, Quadrant II.*
10. **Try It 2:** A. On the y-axis at $(0, 8)$ — *An x-coordinate of 0 means no horizontal move, so the point stays on the y-axis. Moving up 8 places it at $(0, 8)$.*
11. **Exit Ticket:** A. $(6, 3)$ — *Right 6 = x-coordinate of 6, Up 3 = y-coordinate of 3. The ordered pair is $(6, 3)$.*